Т 6

$$p(x) = \begin{cases} \frac{\theta-1}{x^\theta} & x \geq 1 \\ 0 & x < 1 \end{cases} \qquad \theta > 1$$

a) $L(\theta) = \prod_{i=1}^{n} p(x_i, \theta) = \frac{\theta-1}{x_1^\theta} \cdot \frac{\theta-1}{x_2^\theta} \cdot \ldots \cdot \frac{\theta-1}{x_n^\theta} = (\theta-1)^n \frac{1}{(x_1 \cdots x_n)^\theta}$

$$\frac{\partial}{\partial\theta} \ln L(\theta) = \frac{\partial}{\partial\theta}\left(n \ln(\theta-1) - \theta \ln \prod_{i=1}^{n} x_i\right) = 0$$

$$\Rightarrow \theta = 1 + \frac{n}{\ln \prod_{i=1}^{n} x_i}$$

$$(\ln L)''_{\theta\theta} = \frac{-1}{(\theta-1)^2} n < 0 \Rightarrow \text{макс.}$$

$$\hat{\theta} = 1 + \frac{n}{\sum_{i=1}^{n} \ln x_i}$$

b) доверительный интервал для

сильная регулярность:

$p(x, \hat{\theta})$ дважды непр дифф по $\hat{\theta}$ на B

$$\frac{\partial}{\partial\theta} \int_1^\infty p(x,\theta)\,dx = \frac{\partial}{\partial\theta} \int_1^\infty \frac{\theta-1}{x^\theta}\,dx = \frac{\partial}{\partial\theta}\left(\frac{\theta-1}{\theta-1}\right) = 0$$

$$\int_1^\infty \frac{\partial}{\partial\theta} \cdot p(x,\theta)\,dx = \int_1^\infty \frac{\partial}{\partial\theta}\left(\frac{\theta-1}{x^\theta}\right) dx = \int_1^\infty \frac{x^\theta - (\theta-1)x^\theta \ln x}{x^{2\theta}}\,dx =$$

$$= \int_1^\infty \frac{1-(\theta-1)\ln x}{x^\theta}\,dx = \int_1^\infty \frac{1}{x^\theta}\,dx - (\theta-1)\int_1^\infty \frac{\ln x}{x^\theta}\,dx = \frac{x^{1-\theta}}{1-\theta}\Big|_1^\infty - (\theta-1) \cdot$$

$$\cdot \int_1^\infty \frac{\ln x}{x^\theta}\,dx = 0$$

$\Rightarrow$ переставовочная вторых производных

$I(\theta) > 0$

$\Rightarrow$ модель сильно регулярна

x_{med} — медиана

$$\int_1^{x_{med}} p\,dx = \int_1^{x_{med}} \frac{\theta-1}{x^\theta}\,dx = \frac{1}{2} \Rightarrow x_{med} = 2^{\frac{1}{\theta-1}}$$

$$-x^{-\theta+1}\Big|_1^{x_{med}} = \frac{1}{2}$$

$$-x_{med}^{-\theta-1} = -\frac{1}{2}$$

$$g(\hat{\theta}) = 2^{\frac{1}{\hat{\theta}-1}}$$

$$\sqrt{n}\,\frac{g(\hat{\theta}) - g(\theta)}{\sigma(\hat{\theta})} \rightsquigarrow N(0,1), \quad \sigma^2(\theta) = \nabla^T g(\theta)\, I^{-1}(\theta)\, \nabla g(\theta)$$

$$I(\theta) = M\left[\left(\frac{\partial \ln p(?,\theta)}{\partial\theta}\right)^2\right] \overset{\text{т.к. } \ln p(x) = \ln(\theta-1) - \theta \ln x}{=\!=\!=} \int_1^\infty \left(\frac{1}{\theta-1} - \ln x\right)^2 \frac{\theta-1}{x^\theta}\,dx$$

$$I_1 = \int_1^\infty \frac{1}{\theta-1}\,\frac{1}{x^\theta}\,dx = \frac{1}{\theta-1}\int_1^\infty x^{-\theta}\,dx = \frac{x^{-\theta+1}}{-\theta+1}\Big|_1^\infty \frac{1}{\theta-1} = \frac{1}{(\theta-1)^2}$$

$$I_2 = \int_1^\infty \frac{-2\ln x}{\theta-1}\,\frac{\theta-1}{x^\theta}\,dx = -2\int_1^\infty \frac{\ln x}{x^\theta}\,dx = -2\int_1^\infty \ln x\, d\left(x^{-\theta+1}\right)\frac{-1}{-\theta+1} =$$

$$= -2\left[\ln x\left(-\frac{x^{1-\theta}}{\theta-1}\right)\Big|_1^\infty - \int_1^\infty\left(-\frac{x^{1-\theta}}{\theta-1}\right)\frac{1}{x}\,dx\right] = 2\int_1^\infty \frac{x^{-\theta}}{1-\theta} = \frac{-2}{(1-\theta)}$$

$$I_3 = \int_1^\infty \frac{\ln^2 x}{x^\theta}(\theta-1)\,dx = (\theta-1)\int_1^\infty \frac{\ln^2 x}{x^\theta}\,dx = \frac{2}{(\theta-1)^2}$$

$$\Rightarrow \boxed{I(\theta) = \frac{1}{(\theta-1)^2}}$$

$$\nabla g = 2^{\frac{1}{\theta-1}} \cdot \ln 2\,\frac{-1}{(\theta-1)^2}$$

$$\sqrt{n}\,\frac{2^{\frac{1}{\hat\theta-1}} - 2^{\frac{1}{\theta-1}}}{\sqrt{\left(2^{\frac{1}{\theta-1}}\ln 2\,\frac{-1}{(\theta-1)^2}\right)^2\cdot(\hat\theta-1)^2}} \rightsquigarrow N(0,1)$$

$$\sqrt{n}\,\frac{\left(2^{\frac{1}{\hat\theta-1}} - 2^{\frac{1}{\theta-1}}\right)(\hat\theta-1)}{2^{\frac{1}{\theta-1}}\ln 2} \rightsquigarrow N(0,1)$$

$$t_1 < \sqrt{n}\,\frac{\left(2^{\frac{1}{\hat\theta-1}} - 2^{\frac{1}{\theta-1}}\right)(\hat\theta-1)}{2^{\frac{1}{\theta-1}}\ln 2} < t_2$$

$$\frac{\frac{t_1}{\sqrt{n}}\,2^{\frac{1}{\theta-1}}\ln 2}{\hat\theta-1} < g(\hat\theta) - g(\theta) < \frac{\frac{t_2}{\sqrt{n}}\,2^{\frac{1}{\theta-1}}\ln 2}{\hat\theta-1}$$

$$\frac{t_1\,2^{\frac{1}{\theta-1}}\ln 2}{\sqrt{n}(\hat\theta-1)} - g(\hat\theta) < g(\theta) < \frac{t_2\,2^{\frac{1}{\theta-1}}\ln 2}{\sqrt{n}(\hat\theta-1)} - g(\hat\theta)$$

c) асимптот. дов интервал для θ

$$\sqrt{n}\,\frac{\hat\theta-\theta}{\sqrt{(\hat\theta-1)^2}} \rightsquigarrow N(0,1)$$

$$t_1 < \frac{\sqrt{n}(\hat\theta-\theta)}{\hat\theta-1} < t_2$$

$$\frac{t_1}{\sqrt{n}}(\hat\theta-1) - \hat\theta < -\theta < \frac{t_2}{\sqrt{n}}(\hat\theta-1)$$

$$\hat\theta - \frac{t_2(\hat\theta-1)}{\sqrt{n}} < \theta < \hat\theta - \frac{t_1(\hat\theta-1)}{\sqrt{n}}$$